REG-01-PATENT-PORTFOLIO

REG-01: COMPREHENSIVE PATENT PORTFOLIO STRATEGY

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I. EXECUTIVE SUMMARY

Carrington Storm Motors has identified 42 inventions across five technology categories as candidate patent applications. Of these, 19 have been filed as U.S. provisional patent applications, preserving priority dates while providing 12 months to convert to non-provisional or PCT applications. The remaining 23 inventions are documented at the invention disclosure stage and will enter the provisional pipeline as priority dates and resource allocation permit. Total estimated patent prosecution cost across all 42 inventions in all target jurisdictions: $2.5M over 5 years, allocated in Series A budget.

This document outlines: - The five technology categories and the inventions within each - Claims outlines for the 12 highest-priority inventions - Jurisdiction strategy for US, EP, CN, JP, and KR markets - Provisional-to-non-provisional conversion timeline - PCT filing strategy and timeline - Patent landscape analysis - Estimated filing and prosecution costs

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II. TECHNOLOGY CATEGORIES AND INVENTION COUNTS

Category A: MXene Compositions and Synthesis (12 Inventions)

These inventions cover the composition of matter and methods of synthesis for CSM's proprietary MXene materials.

A-1: "Dielectric Polymer-MXene Composite for Electromagnetic Interference Shielding" - Provisional Filed: May 15, 2026 (US 63/XXX,XXX) - Priority Date: May 15, 2026 - Abstract: A composite material comprising exfoliated MXene nanosheets dispersed in a dielectric polymer matrix, wherein the MXene loading is between 5% and 25% by weight, achieving electromagnetic interference shielding effectiveness of 60-75 dB in the frequency range of 0.1 MHz to 10 GHz at a composite thickness of 0.3-1.0 mm. The invention further includes a method of manufacturing the composite via melt-extrusion processing with precisely controlled shear rates to achieve planar alignment of MXene nanosheets, maximizing shielding effectiveness per unit thickness.

Claims Outline: - Claim 1 (Independent): A composite material comprising a dielectric polymer matrix and exfoliated MXene nanosheets dispersed therein, wherein the MXene nanosheets comprise a transition metal carbide or carbonitride having the formula $M_{(n+1)}X_nT_x$, where M is at least one transition metal, X is carbon and/or nitrogen, n = 1, 2, or 3, and T_x represents surface termination groups selected from -O, -OH, and -F. - Claim 5 (Dependent): The composite of claim 1, wherein the MXene loading is 5-25 wt%. - Claim 8 (Dependent): The composite of claim 1, wherein the EMI SE is 60-75 dB at 0.1 MHz-10 GHz. - Claim 12 (Dependent): The composite of claim 1, wherein the dielectric polymer is selected from polyethylene, polypropylene, polyvinylidene fluoride, and copolymers thereof. - Claim 15 (Independent Method): A method of manufacturing the composite of claim 1, comprising the steps of melt-extruding a mixture of the dielectric polymer and MXene nanosheets at a controlled shear rate of 50-500 s^{-1} to achieve planar alignment. - Claim 20 (Dependent): The method of claim 15, wherein the extrusion is performed at a temperature between 180°C and 280°C. - Additional claims: 28 total (including dependent claims covering specific MXene stoichiometries, polymer blends, multilayer constructions, and field-applicable wrapping configurations).

Priority Date Strategy: Provisional filed May 15, 2026. Non-provisional and PCT conversion deadline: May 15, 2027.

Jurisdiction Strategy: US, EP, CN, JP, KR. CN is critical because China is the primary source of titanium precursors for MXene synthesis. JP and KR are critical because of advanced electronics manufacturing presence and strong IP enforcement regimes.

Estimated Filing Cost: $28,000 (US non-provisional filing, search, and examination fees) + $5,500 per additional jurisdiction (PCT national phase entry) = $50,000 for all five jurisdictions.

Commercialization Pathway: Direct use in ShieldCore transformer housing, ConductorCore transmission line wrap, DriveShield vehicle kit, and FieldPak communications shield.

A-2: "Method for Continuous-Flow Exfoliation of MXene Nanosheets" - Provisional Filed: June 22, 2026 (US 63/XXX,XXX) - Priority Date: June 22, 2026 - Abstract: A continuous-flow process for exfoliating MAX phase precursors to produce MXene nanosheets, comprising: (a) continuously feeding a MAX phase powder into an etching solution under controlled temperature and flow rate; (b) applying ultrasonic energy in a flow-through sonication cell to delaminate the etched MAX phase into individual MXene nanosheets; (c) separating exfoliated MXene nanosheets from unexfoliated material via differential centrifugation; and (d) recycling unexfoliated material back to the sonication step. The process achieves MXene yield exceeding 85% compared to approximately 40% for conventional batch processes.

Claims Outline: - Claim 1 (Independent Method): A method for producing MXene nanosheets, comprising the steps of [continuous flow parameters, etching conditions, sonication parameters]. - Claim 10 (Dependent): The method of claim 1, wherein the yield of exfoliated MXene nanosheets exceeds 80% of the MAX phase precursor by weight. - Claim 15 (Dependent): The method of claim 1, further comprising recycling unexfoliated MAX phase material from the separation step to the sonication step. - 22 total claims.

Priority Date Strategy: Provisional filed June 22, 2026. Non-provisional and PCT conversion deadline: June 22, 2027.

Jurisdiction Strategy: US, EP, CN, JP, KR. Same rationale as A-1, with particular emphasis on CN for manufacturing process patent enforcement.

Estimated Filing Cost: $28,000 (US) + $5,500 per additional jurisdiction = $50,000.

A-3: "MXene-Polymer Multilayer Composite with Gradient Composition for Broadband Electromagnetic Shielding" - Invention Disclosure Stage (Provisional Filing Planned: Q3 2026) - Abstract: A multilayer composite construction comprising at least three layers of MXene-polymer composite, each layer having a distinct MXene loading concentration, arranged in a gradient from high MXene loading (outermost layer, 15-25 wt%) to low MXene loading (innermost layer, 5-10 wt%), achieving broadband EMI shielding from 100 kHz to 18 GHz with absorption-dominated shielding mechanism. - Estimated 25 claims.

A-4 through A-12 (Summarized): - A-4: "Surface-Functionalized MXene Nanosheets with Enhanced Polymer Compatibility" - A-5: "MXene-Carbon Nanotube Hybrid Composite for Synergistic Electromagnetic Shielding" - A-6: "Self-Healing MXene-Polymer Composite for Extended Service Life" - A-7: "MXene Aerogel with Ultra-Low Density for Weight-Critical Shielding Applications" - A-8: "Transparent MXene Thin Film for Optical Electromagnetic Shielding" - A-9: "MXene-Based Conductive Ink for Additively Manufactured Shielding Components" - A-10: "Aqueous MXene Dispersion with Extended Shelf Stability" - A-11: "MXene-Graphene Hybrid for Enhanced Thermal and Electrical Conductivity" - A-12: "Method for In-Situ MXene Synthesis Within a Polymer Matrix"

Category B: Dielectric Composite Structural Materials (8 Inventions)

These inventions cover structural materials with integrated dielectric properties, primarily for transformer housing and substation infrastructure.

B-1: "Ceramic Matrix Composite Electrical Insulator with Structural Load-Bearing Capacity" - Provisional Filed: July 8, 2026 (US 63/XXX,XXX) - Priority Date: July 8, 2026 - Abstract: A structural composite material comprising silicon carbide fibers embedded in an aluminum oxide ceramic matrix, wherein the volume fraction of SiC fibers is 25-45%, the composite has an electrical resistivity of at least 10^11 ohm-centimeters, a compressive strength of at least 400 MPa, and a thermal conductivity of at least 30 W/m·K. The composite is specifically designed for use as a transformer housing that simultaneously provides structural support, electrical insulation, and passive thermal management.

Claims Outline: - Claim 1 (Independent): A structural composite comprising SiC fibers in an Al_2O_3 matrix, wherein the electrical resistivity is $\geq 10^{11}$ Ω·cm and compressive strength ≥ 400 MPa. - Claim 8 (Dependent):

The composite of claim 1, formed into a transformer housing having wall thickness 15-45 mm. - Claim 14 (Independent Method): A method of manufacturing the composite of claim 1 via chemical vapor infiltration. - 24 total claims.

Jurisdiction Strategy: US, EP, CN, JP, KR. Transformer manufacturing is concentrated in these jurisdictions (ABB/Siemens/Hitachi Energy in Europe and Asia, GE/SPX/Waukesha in North America).

Estimated Filing Cost: \$30,000 (US) + \$6,000 per additional = \$54,000.

B-2 through B-8 (Summarized): - B-2: “Geopolymer Concrete Composition with High Electrical Resistivity for Electrical Substation Foundations” - B-3: “Fiber-Reinforced Geopolymer Composite with Integrated Ground Current Interruption Layer” - B-4: “Method for In-Situ Casting of Dielectric Foundation Systems for Existing Substations” - B-5: “Multi-Layer Dielectric Building Panel with Integrated MXene Shielding Mesh” - B-6: “Precast Dielectric Concrete Module for Modular Substation Construction” - B-7: “Lightweight Dielectric Aggregate for Reduced-Density Electrical Insulation Concrete” - B-8: “Accelerated-Cure Geopolymer Formulation for Field-Deployable Substation Foundation Repair”

Category C: Amphibious Vehicle Mechanisms (7 Inventions)

These inventions cover the mechanical systems of the Sentinel-Class amphibious walker and related maritime vehicles.

C-1: “Quadrupedal Walking Vehicle with Dielectric Hull for Operation in Electromagnetically Hostile Environments” - Provisional Filed: August 12, 2026 (planned) - Abstract: An amphibious walking vehicle comprising a fully sealed dielectric hull constructed of ceramic-matrix composite material, four articulated legs with hydraulic actuation, each leg terminating in an interchangeable foot assembly selectable between a wheel, a tracked pod, and a static pad, wherein all electronic control systems are enclosed within MXene-polymer shielded compartments, enabling operation in environments with electromagnetic field strengths exceeding 50 kV/m without electronic system degradation.

Claims Outline: - Claim 1 (Independent): An amphibious walking vehicle comprising: a hull constructed of material having electrical resistivity exceeding 10^{11} Ω·cm; four legs; hydraulic actuation system; and shielded electronic compartments. - Claim 12 (Dependent): The vehicle of claim 1, wherein the hull material is a SiC-fiber-reinforced Al_2O_3 ceramic matrix composite. - Claim 18 (Dependent): The vehicle of claim 1, wherein the electronic compartments are shielded by MXene-polymer composite providing ≥ 60 dB EMI SE. - 22 total claims.

C-2 through C-7 (Summarized): - C-2: “Interchangeable Foot Assembly for Amphibious Walking Vehicle” - C-3: “Hydraulic Actuation System with Passive Lock for Extended-Duration Static Positioning” - C-4: “Dielectric Hull Sealing System for Amphibious Vehicle Watertight and EM-Tight Integrity” - C-5: “Method for Deploying Grid Protection Equipment Via Amphibious Walking Vehicle” - C-6: “Autonomous Navigation System for Amphibious Vehicle in Debris-Field Environments” - C-7: “Modular Payload Bay for Multi-Mission Amphibious Vehicle Configuration”

Category D: Protonic Communication Systems (6 Inventions)

These inventions cover CSM’s backup communication technology using proton-based signal propagation.

D-1: “Proton-Based Signal Transmission System for Electromagnetic-Independent Communications” - Provisional Filed: September 5, 2026 (planned) - Abstract: A communications system comprising a proton source, a proton modulator configured to encode information onto a proton beam via modulation of proton flux density, a proton detector array disposed at a receiving location, and a decoder configured to extract the encoded information from detected proton flux variations. The system operates independently of electromagnetic signal propagation, enabling communication during geomagnetic storms, EMP events, and other scenarios where electromagnetic communications are disrupted.

Claims Outline: - Claim 1 (Independent): A communication system comprising a proton source, proton modulator, proton detector array, and signal decoder. - Claim 8 (Dependent): The system of claim 1, wherein the proton source is a proton-exchange membrane electrolyzer. - Claim 15 (Dependent): The system of claim 1, wherein the data rate is between 100 bps and 10 kbps. - 20 total claims.

D-2 through D-6 (Summarized): - D-2: “Portable Protonic Transceiver Unit for Field Emergency Communications” - D-3: “Relay-Drone-Deployed Protonic Communication Mesh Network” - D-4: “Proton Beam

Steering Apparatus for Non-Line-of-Sight Protonic Communications” - D-5: “Error-Correction Protocol Optimized for Low-SNR Protonic Signal Channels” - D-6: “Hybrid Electromagnetic-Protonic Communication System with Automatic Failover”

Category E: Smart Materials and Indicators (9 Inventions)

These inventions cover YInMn Blue smart coatings and related passive monitoring technologies.

E-1: “Temperature-Indicating Coating Comprising Doped YInMn Blue Pigment” - Provisional Filed: October 18, 2026 (planned) - Abstract: A thermochromic coating composition comprising YInMn Blue pigment doped with at least one rare earth element at a dopant concentration between 0.5% and 5% by weight, wherein the coating exhibits a reversible color shift from blue to violet at a threshold temperature between 40°C and 120°C, the threshold temperature being determined by the dopant identity and concentration. The coating is applied to electrical equipment as a passive, zero-power thermal monitoring indicator.

Claims Outline: - Claim 1 (Independent): A thermochromic coating composition comprising: YInMn Blue pigment particles doped with a rare earth element at 0.5-5 wt%, dispersed in a binder. - Claim 6 (Dependent): The composition of claim 1, wherein the rare earth element is selected from Eu, Tb, Dy, Ho, Er, Tm, and Yb. - Claim 10 (Dependent): The composition of claim 1, wherein the threshold temperature is tuned by selecting the dopant concentration to achieve a desired transition temperature. - 25 total claims.

E-2 through E-9 (Summarized): - E-2: “Multi-Threshold Temperature-Indicating Coating System With Distinct Color Transitions” - E-3: “Substrate-Specific Primer for Adhesion of YInMn Blue Smart Coating to Ceramic Surfaces” - E-4: “Accelerated-Weathering YInMn Blue Coating with UV-Stabilized Binder Matrix” - E-5: “Method for Interpreting YInMn Blue Color Shift for Asset Health Monitoring” - E-6: “YInMn Blue-Coated Transformer Bushing Inspection System” - E-7: “Color-Shift Calibration Standard for Quantitative YInMn Blue Coating Interpretation” - E-8: “Multi-Spectral Imaging System for Automated YInMn Blue Coating Inspection” - E-9: “YInMn Blue Doping Method Using Solid-State Synthesis with Controlled Atmosphere”

III. PATENT LANDSCAPE ANALYSIS

A. Key Third-Party Patents and Prior Art

Drexel University MXene Patent Family (Yury Gogotsi / Michel Barsoum): - US 9,418,498: “Two-dimensional, ordered, two-dimensionally corrugated transition metal carbides and nitrides” (priority 2012) - US 9,752,982: “Electrochemical devices comprising MXene electrodes” (priority 2013) - US 10,217,569: “Electromagnetic interference shielding comprising MXene” (priority 2017) - Scope: Covers MXene compositions generally and EMI shielding applications. Our MXene-polymer composite formulations are designed to be distinguishably different from the Drexel compositions in specific polymer matrices, MXene loading ranges, and processing methods. We have freedom to operate with respect to these patents through composition-of-matter differentiation and method-of-manufacture differentiation. However, we are closely monitoring the Drexel portfolio for any new filings that could create conflicts.

Boeing Composite Patents: - US 8,309,644: “Ceramic composite with integrated compliance layer” (priority 2008) - US 9,828,276: “Ceramic matrix composite turbine engine component” (priority 2015) - Scope: Covers ceramic matrix composites for aerospace applications. Our CMC transformer housing is in a different technical field (electrical infrastructure vs. aerospace propulsion) and uses different fiber-matrix combinations and processing conditions. Freedom to operate is strong.

Existing EMP Shielding Patents: - Multiple patents exist for Faraday cage enclosures, surge suppression devices, and EMP-hardened electronics (e.g., US 6,324,075, US 7,684,165, US 8,928,328). - CSM’s approach is fundamentally different: rather than enclosing individual components in conductive cages, we modify the materials of the infrastructure itself to be non-conductive at the bulk level while providing EMI shielding at the surface level. This materials-science-based approach avoids the enclosures that are the subject of existing EMP shielding patents.

B. Patent Landscape Map

Technology Area	Patent Density	Dominant Holders	CSM Positioning
MXene synthesis and	Medium (50-80 active	Drexel, Murata, Samsung	Differentiated polymer composites and

compositions	families)		manufacturing methods
EMI shielding materials	High (200+ active families)	3M, Laird, Parker Hannifin	MXene-based approach is novel; most prior art is metal mesh or conductive polymer
Ceramic matrix composites	High (300+ families)	GE, Rolls-Royce, Boeing	Electrical infrastructure application is novel; prior art is aerospace propulsion
Geopolymer concrete	Low (20-30 active families)	Various universities	Dielectric application is novel; most prior art is structural or environmental
Thermochromic materials	Medium (60-90 families)	Various	YInMn Blue doping for specific threshold temperatures is novel
Amphibious vehicles	Medium-High (150+ families)	BAE Systems, Lockheed, Gibbs	Dielectric hull + quadrupedal walking + utility deployment is a novel combination
Proton-based communications	Very Low (<10 families)	Various research groups	This is a frontier technology area with minimal patent thicket

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IV. PROVISIONAL VS. NON-PROVISIONAL STRATEGY

A. Strategic Rationale for Provisional Filings

We file provisionally for the following reasons: 1. **Priority Date Preservation:** Establishing the earliest possible priority date for each invention. 2. **Cost Deferral:** Provisional filing costs ($140 micro-entity, $280 small entity) are approximately 1/100th of non-provisional filing costs. With 42 inventions, filing all as non-provisionals immediately would cost approximately $1.2M in filing fees alone, before prosecution costs. 3. **12-Month Development Window:** The provisional period allows us to generate additional supporting data, refine claims, and assess commercial viability before committing to full prosecution. 4. **Stealth Period:** Provisionals are not published. They provide priority protection while maintaining trade secrecy until the non-provisional publishes at 18 months.

B. Conversion Decision Criteria

Each provisional will be evaluated for non-provisional conversion based on: 1. **Commercial Viability:** Is the product incorporating this invention on the near-term commercialization path? 2. **Competitive Intelligence:** Is there evidence that competitors are working on similar approaches? 3. **Data Sufficiency:** Is the experimental data sufficient to support the claimed scope? 4. **Freedom-to-Operate Value:** Does having this patent strengthen our position against potential infringement claims? 5. **Licensing Potential:** Could this patent generate licensing revenue independently?

Inventions that do not meet the conversion threshold will be defensively published (see REG-02: Trade Secret Strategy) to block competitors.

V. PCT TIMELINE

The Patent Cooperation Treaty (PCT) provides a unified international filing mechanism that preserves the right to file in 157 member countries for 30-31 months from the priority date.

PCT Filing Timeline for CSM

Priority Date	PCT Deadline (12 months)	National Phase Deadline (30-31 months)
May 15, 2026 (A-1)	May 15, 2027	November 15, 2028
June 22, 2026 (A-2)	June 22, 2027	December 22, 2028
July 8, 2026 (B-1)	July 8, 2027	January 8, 2029

PCT Strategy

For the 12 highest-priority inventions, we will file a single PCT application (claiming priority to the relevant provisional) covering all five target jurisdictions (US, EP, CN, JP, KR). This provides: - Unified search and preliminary examination - Extended timeline (30-31 months) for national phase entry - Deferred translation costs (translation only needed at national phase entry, not at PCT filing) - International Search Report providing early feedback on patentability before major prosecution investment

Estimated PCT filing cost per invention: $5,500 (transmittal, search, and international fees for small entity) + $3,000 (attorney fees for PCT preparation from provisional) = $8,500 per invention.

VI. JURISDICTION STRATEGY

A. Jurisdiction Selection Rationale

Jurisdiction	Rationale	Cost Per Invention (National Phase)
United States (USPTO)	Home jurisdiction, largest target market, strongest enforcement	$5,500
European Patent Office (EPO)	Single application covers 39 member states; key market for Siemens/ABB/Eaton	$8,000 (incl. validation in DE, FR, UK)
China (CNIPA)	MXene precursor supply chain; growing grid infrastructure market; requires local CN filing for enforcement	$4,500
Japan (JPO)	Advanced electronics and materials market; strong IP enforcement; Toshiba/Hitachi/Mitsubishi presence	$5,000
South Korea (KIPO)	Advanced manufacturing; Samsung/LG materials R&D; strong IP enforcement	$4,500

B. Tier-2 Jurisdictions (Deferred Consideration)

Jurisdiction	Rationale	Deferral Trigger
Australia	Significant mining and grid infrastructure; strong IP system	Customer traction in AU/NZ
Canada	Grid interconnection with US; Hydro One is potential customer	Commercial contract in CA
India	Large and growing grid; significant solar storm exposure at low latitudes	Regulatory engagement in IN
Brazil	Large grid; significant GIC risk at low latitudes; INMETRO certification pathway	Customer or manufacturing presence in BR

VII. COST ESTIMATE

A. Filing Costs (Years 1-3)

Category	Inventions	US Non-Provisional	PCT	National Phase (5 jurisdictions)	Total Per Invention
High-Priority (A-1 through E-1)	12	$28,000 × 12 = $336,000	$8,500 × 12 = $102,000	$27,000 × 12 = $324,000	$762,000
Medium-Priority	18	$22,000 × 18 = $396,000	$≥7,000 × 18 = $126,000	Deferred	$522,000
Low-Priority	12	Defensive publication	N/A	N/A	$18,000

B. Prosecution Costs (Years 1-5)

Estimated office action responses: 2-3 per invention in US, 1-2 in other jurisdictions. - US prosecution per invention: $8,000-15,000 (attorney fees for office action responses) - EP prosecution per invention: $6,000-12,000 - CN/JP/KR prosecution per invention: $4,000-8,000 each - Total prosecution cost (30 inventions × average $25,000): $750,000

C. Maintenance Fees (Years 3-20)

- US maintenance fees: $2,000 (3.5 yr), $3,760 (7.5 yr), $7,700 (11.5 yr) per patent
- EP maintenance fees: Vary by country; average $1,500/yr per validated country
- Total maintenance cost over patent life for 30 issued patents in 5 jurisdictions: approximately $800,000

D. Total IP Budget

Phase	Timeline	Cost
Provisional filings (19 filed, 23 to file)	2026-2027	$15,000
Non-provisional and PCT filings	2027-2028	$984,000
National phase entries	2028-2029	$450,000
Prosecution	2027-2031	$750,000
Defensive publications	2026-2028	$18,000
Maintenance fees	2029-2046	$800,000
Total (Series A allocation)	**2026-2029**	**$2,500,000**
Total (lifetime)	**2026-2046**	**$3,017,000**

The $2.5M allocated in the Series A budget covers the first four years of IP activity: all provisional and non-provisional filings, PCT applications, national phase entries for the 12 high-priority inventions, initial prosecution, and defensive publications. Later-phase prosecution and maintenance costs will be funded from operating revenue or follow-on financing.

VIII. PATENT PROSECUTION TIMELINE (GANTT-STYLE NARRATIVE)

July 2026: 19 provisional patents filed. Begin drafting additional 23 provisional applications. **September 2026:** Engage IP counsel of record (RFP process complete; 3 candidate firms identified). **November 2026:** Complete filing of all 42 provisional applications. **January 2027:** Initiate prior art searches for high-priority inventions. **May 2027:** File non-provisional and PCT applications for first group (A-1, A-2, related filings). First priority date anniversaries. **June-July 2027:** File non-provisional and PCT applications for second group (B-1, related). **August-September 2027:** File non-provisional and PCT for third group (C-1, related). **October 2027:** File non-provisional and PCT for fourth group (D-1, related). **November-December 2027:** File non-provisional and PCT for fifth group (E-1, related). **January-December 2028:** PCT International Search Reports received. Begin national phase preparations. Respond to first US office actions. **November 2028:** First PCT national phase entry deadline (November 15, 2028 for May 15, 2026 priority). **2029:** All remaining national phase entries. Continuation and divisional filings as needed. **2030-2031:** Substantive examination in all jurisdictions. Patent grants begin.

IX. PATENT-TO-PRODUCT MAPPING (High-Priority Inventions Only)

Patent Family	Primary Product(s)	Commercialization Timeline
A-1 (MXene-Polymer Composite)	ShieldCore, ConductorCore, DriveShield, FieldPak, HomeShield	2028-2030
A-2 (Continuous-Flow Exfoliation)	Internal manufacturing process	2027-2028
A-3 (Gradient Composite)	ShieldCore Gen 2	2030+
B-1 (CMC Transformer Housing)	ShieldCore	2028-2030
B-2 (Geopolymer Concrete)	SubShield Foundation	2027-2029
C-1 (Amphibious Walker)	Sentinel-Class	2031+
C-4 (Dielectric Hull Seal)	Sentinel-Class, Resilience-Class	2031+
D-1 (Protonic Transceiver)	Protonic Comms Network, RelayDrone	2029-2031
E-1 (YInMn Blue Coating)	All shielded products (aftermarket)	2028-2030
E-2 (Multi-Threshold Coating)	Gen 2 smart coating applications	2031+

X. RISKS AND MITIGATIONS

Risk	Probability	Impact	Mitigation
Drexel MXene patent blocks our composition claims	Medium	High	Design-around formulations with distinct polymer matrices and processing conditions; licensing negotiation as backup
Provisional filing is insufficient to support full non-provisional claims	Low	Medium	Over-document in provisionals; include extensive experimental data; file additional provisionals with supplementary data before conversion
Competitor files overlapping claims before our priority dates	Low	High	Accelerate provisional filing schedule; conduct regular prior art monitoring
PCT international search report reveals unpatentable subject matter	Medium	Medium	Narrow claims to patentable scope; identify fallback claim strategies in each provision
Insufficient resources to prosecute all 42 inventions effectively	Medium	Medium	Triage inventions by commercial priority; defensively publish low-priority rather than abandon
Trade secret vs. patent tension — filing a patent requires disclosure of the invention	N/A (inherent)	Medium	Patents cover compositions and devices; trade secrets cover synthesis parameters and exact processing conditions; no single document contains both

APPENDIX: PATENT NUMBERING SYSTEM

CSM uses the following internal patent reference system:

Format: CSM-P[Category]-[Sequence Number]-[Year] - Category letters: A (MXene), B (Dielectric Composites), C (Amphibious), D (Protonic), E (Smart Materials) - Example: CSM-PA-001-26 = MXene category, invention 001, filed in 2026

This system is used in all internal documentation and correspondence with IP counsel. Provisional serial numbers are mapped to internal references in a confidential cross-reference maintained by Rook (Security Director).

END OF DOCUMENT — CSM-REG-01-v2.3